

Math 290

Practice Quiz 2 — Version A

(extra practice, Sections 6–11)

1. Short response.

(a) Evaluate the following proposed theorem and proof.

Theorem: Let $a, b \in \mathbb{Z}$. If $a \mid 2b$, then $a \mid b$.

Proof. Let $a, b \in \mathbb{Z}$. We work directly. Assume $a \mid 2b$. Thus $2b = ac$ for some $c \in \mathbb{Z}$. Now

$$b = \frac{ac}{2} = a \left(\frac{c}{2} \right).$$

Since $c/2 \in \mathbb{Z}$, we have $a \mid b$. □

A. The theorem is true and the proof is correct.

B. The theorem is false.

C. The theorem is true, but the proof is incorrect because it should have been done contrapositively.

D. The theorem is true, but the proof is incorrect because it never uses the fact that b is an integer.

E. The theorem is true, but the proof is incorrect because it divides by a .

(b) True or false: The statement “Let $x \in \mathbb{R}$. If $x^2 < 0$, then $x = 5$.” is vacuously true.

(c) True or false: $2 \equiv -2 \pmod{3}$.

2. Let $x, y \in \mathbb{Z}$. Prove that if $xy + y^2$ is even, then x is odd or y is even.

3. Prove that $\sqrt{3}$ is irrational.

4. Prove the following statement (giving a *single* justification for every step you make).
For any two sets X and Y , we have $X - (X - Y) = X \cap Y$.